Tech Dessert #5: Making Contacts and Using Repeaters

Companion reference for meeting 5 -- how to call and answer other stations, and how repeaters, tones, and Q signals work.

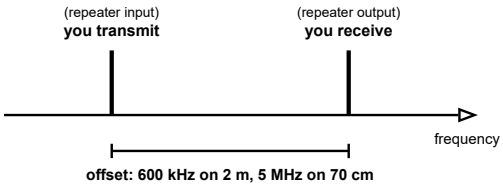
Groups T2A + T2B · 25 pool questions · Self-study handout · Practice exams at eham.net/exams

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This Dessert is a reference sheet – the practical habits of getting on the air. Read the tables, study the repeater-offset diagram, then drill the question bank until the answers come automatically.

Making Contacts

Topic	What to know	Q
What “CQ” means	**Calling any station** – an invitation for anyone to answer.	T2A08
How to answer a station calling CQ	Send **the other station’s call sign, then your own**.	T2A05
Calling CQ yourself (not on a repeater)	Say “CQ” a few times, then “this is” and your call sign; pause and listen.	T2A06
Calling a specific station on a repeater	Say that station’s call sign, then identify with yours.	T2A04
Showing you are listening on a repeater	Your call sign followed by the word **“listening.”**	T2A09
2-meter FM simplex calling frequency	**146.520 MHz.**	T2A02
What “simplex” means	Transmitting and receiving on the **same frequency**.	T2A11
What a “repeater offset” is	The difference between a repeater’s **transmit and receive** frequencies.	T2A07
Common 2-meter repeater offset	Plus or minus **600 kHz**.	T2A01
Common 70-centimeter repeater offset	Plus or minus **5 MHz**.	T2A03
What a “band plan” is	A **voluntary guideline** for which modes and activities go where in a band.	T2A10



A repeater receives (its input) and transmits (its output) on two different frequencies. Your radio’s “offset” sets that gap automatically.

Using Repeaters, Tones, and Q Signals

Topic	What to know	Q
Purpose of the squelch function	**Mutes the receiver audio** when no signal is present.	T2B13
Sub-audible tone that opens a receiver’s squelch	**CTCSS.**	T2B02
Signaling that uses two simultaneous audio tones	**DTMF.**	T2B06
The “reverse” function	Lets you listen on the repeater’s **input** frequency.	T2B01
You can hear a repeater but cannot access it	Wrong CTCSS tone, wrong DCS code, or improper offset – **any of these**.	T2B04
Your FM audio drops out on voice peaks	You are **talking too loudly**.	T2B05
A linked repeater network	Signals received by one repeater are retransmitted by **all repeaters** in the network.	T2B03
Why simplex channels are in band plans	So nearby stations can talk **without tying up a repeater**.	T2B09
Two stations interfering on one frequency	They should **negotiate** continued use of the frequency.	T2B08
The Q signal QRM	You are **receiving interference** from other stations.	T2B10
The Q signal QSY	You are **changing frequency**.	T2B11
A DMR “talkgroup”	A DMR identifier that organizes traffic so listeners hear **only their group**.	T2B14

Joining a digital repeater's talkgroup	**Program your radio** with the group's ID or code.	T2B07
A DMR "color code"	An access code programmed into a DMR transmitter to reach a **specific repeater**.	T2B12

Remember: to answer a station, send **their call, then yours**. A **repeater** listens and transmits on two frequencies – the **offset** (600 kHz on 2 m, 5 MHz on 70 cm) sets the gap. If you can hear a repeater but cannot key it up, suspect the wrong **CTCSS tone** or offset. And **simplex** means both stations on one frequency, no repeater involved.

Dessert 5 Question Bank (T2A + T2B)

T2A01. What is a common repeater frequency offset in the 2-meter band?
A. Plus or minus 5 MHz
B. Plus or minus 600 kHz
C. Plus or minus 500 kHz
D. Plus or minus 1 MHz

T2A02. What is the national calling frequency for FM simplex operations in the 2-meter band?
A. 146.520 MHz
B. 145.000 MHz
C. 432.100 MHz
D. 446.000 MHz

T2A03. What is a common repeater frequency offset in the 70-centimeter band?
A. Plus or minus 5 MHz
B. Plus or minus 600 kHz
C. Plus or minus 500 kHz
D. Plus or minus 1 MHz

T2A04. What is an appropriate way to call another station on a repeater if you know the other station's call sign?
A. Say "break, break," then the other station's call sign, followed by your call sign
B. Say the station's call sign, then identify with your call sign
C. Say "CQ" three times, then the other station's call sign, followed by your call sign
D. Wait for the station to call CQ, then answer with your call sign

T2A05. How should you respond to a station calling CQ?
A. Transmit "CQ" followed by the other station's call sign
B. Transmit your call sign followed by the other station's call sign
C. Transmit the other station's call sign followed by your call sign
D. Transmit a signal report followed by your call sign

T2A06. What is an effective way to seek a call from any phone station when not using a repeater?
A. Sign your call sign once, followed by the words "listening for a call;" if there is no answer, change the frequency and repeat
B. Say "QTC," followed by "this is" and your call sign; if there is no answer, change the frequency and repeat
C. Transmit an unmodulated carrier for approximately 10 seconds, followed by "this is" and your call sign, and pause to listen; repeat as necessary
D. Repeat "CQ" a few times, followed by "this is," and your call sign, then pause to listen; repeat as necessary

T2A07. What does the term "repeater offset" mean?
A. The difference between a repeater's transmit and receive frequencies
B. The repeater has a time delay to prevent interference
C. The approximately half-second delay to allow for the "squelch tail" to be removed
D. The fee charged by the repeater owner or club to provide for maintenance

T2A08. What is the meaning of the procedural signal "CQ"?
A. A shortened form of the term "Contest QSO"
B. "Communication Quality," used to indicate transmitted audio clarity
C. Only the called station should transmit
D. Calling any station

T2A09. Which of the following is a customary way to indicate a station is listening on a repeater and looking for a contact?
A. "CQ CQ" followed by the repeater's call sign
B. The station's call sign followed by the word "listening"
C. The repeater's call sign followed by the station's call sign
D. "QSY" followed by your call sign

T2A10. What is a band plan, beyond the privileges established by the FCC?
A. A voluntary guideline for using different modes or activities within an amateur band
B. A list of operating schedules
C. A list of available net frequencies
D. A plan devised by a club to indicate frequency band usage

T2A11. What term describes an amateur station that is transmitting and receiving on the same frequency?
A. Full duplex
B. Diplex
C. Simplex
D. Multiplex

T2B01. What is the purpose of the reverse function on a VHF/UHF transceiver?
A. To reduce power output
B. To increase power output
C. To listen on a repeater's input frequency
D. To listen on a repeater's output frequency

T2B02. What term describes the use of a sub-audible tone transmitted along with normal voice audio to open the squelch of a receiver?
A. Carrier squelch
B. Tone burst
C. DTMF
D. CTCSS

T2B03. Which of the following describes a linked repeater network?
A. A network of repeaters in which signals received by one repeater are transmitted by all the repeaters in the network
B. A single repeater with more than one receiver
C. Multiple repeaters with the same control operator
D. A system of repeaters linked by APRS

T2B04. Which of the following could be the reason you are unable to access a repeater whose output you can hear?
A. Improper transceiver offset
B. You are using the wrong CTCSS tone
C. You are using the wrong DCS code
D. All these choices are correct

T2B05. Which of the following would cause your FM transmission audio to drop out on voice peaks?
A. Your repeater offset is inverted
B. Your FM deviation is too low
C. You are talking too loudly
D. Your transmit power is too high

T2B06. What type of signaling to a repeater uses two simultaneous audio tones?
A. DTMF
B. CTCSS
C. GMRS
D. D-STAR

T2B07. How can you join a digital repeater's "talkgroup"?
A. Register with the local Frequency Coordinator
B. Register with the digital repeater control operator
C. Program your radio with the group's ID or code
D. Program your radio with the DTMF tone used by the talkgroup

T2B08. Which of the following applies when two stations transmitting on the same frequency interfere with each other?
A. The stations should negotiate continued use of the frequency
B. Both stations should choose another frequency to avoid conflict
C. Whichever station was on the frequency first has preemptive rights to the frequency
D. Use subaudible tones so both stations can share the frequency

T2B09. Why are simplex channels designated in the VHF/UHF band plans?
A. So stations within range of each other can communicate without tying up a repeater
B. For contest operation
C. For working DX only
D. So stations with simple transmitters can access the repeater without automated offset

T2B10. Which Q signal indicates that you are receiving interference from other stations?
A. QRM
B. QRN
C. QTH
D. QSB

T2B11. Which Q signal indicates that you are changing frequency?
A. QRU
B. QSY
C. QSL
D. QRZ

T2B12. What is the digital color code used on DMR repeater systems?
A. An access code which must be programmed into a DMR transmitter to access a specific repeater
B. A code which automatically programs the repeater offset to a frequency chosen by the user
C. A code which identifies the specific CODEC being used so that the repeater can properly decode the audio
D. A code transmitted by the repeater to indicate health and status of the equipment

T2B13. What is the purpose of a squelch function?
A. Reduce a CW transmitter's key clicks
B. Mute the receiver audio when a signal is not present
C. Eliminate parasitic oscillations in an RF amplifier
D. Reduce interference from impulse noise

T2B14. Which of the following is a "talkgroup"?
A. A group of amateur radio operators who share email on a specific amateur radio topic
B. An identifier used by DMR to organize radio traffic so that those who want to hear the group aren't bothered by other radio traffic
C. The members of a net organized for amateur radio operators with a specific common interest
D. A method of automatically alerting all group members to the frequency where any member is transmitting

Answer Key

T2A01: B	T2A02: A	T2A03: A	T2A04: B	T2A05: C
T2A06: D	T2A07: A	T2A08: D	T2A09: B	T2A10: A
T2A11: C	T2B01: C	T2B02: D	T2B03: A	T2B04: D
T2B05: C	T2B06: A	T2B07: C	T2B08: A	T2B09: A
T2B10: A	T2B11: B	T2B12: A	T2B13: B	T2B14: B